

PIT WALL – Brutal Engineering Risk Audit

Project: AI Race Engineer for disabled/veteran/grassroots racers
(IBM SkillsBuild, May 2026)

Target: 60-second live demo on a laptop CPU, deployed on
Hugging Face Spaces (free tier)

1. Latency Reality Check

Pipeline Step	Tool / Model	Hardware	Realistic Latency (seconds)	Source / Rationale
Parse COA PDF (6-8 pages)	Granite-Docling 258M	Laptop CPU	600-900 s (10-15 min)	A simple PDF took 399 s on an RTX 5090; CPU will be

				2-3× slo wer.
Docling library overhead	Docling pipeline	CPU	5-10 s	I/O + layout detection add minor overhead.
Parse timing-sheet PDF (multi-column)	Granite Vision 4 .1 4B	CPU	120-300 s (2-5 min)	4B VLM; rendering + inference per page ≈30-60 s on CPU.
Zero-shot forecast (6 k rows × 8 channels)	Granite-TimeSe ries-TTM-r2.1	CPU	5-15 s	IBM clai ms “few sec ond s” for zero -sho t; 6 k row s

				may pus h it high er.
Generate coaching report (~300 words)	Granite 4.1 8B I nstruct	CPU	40-80 s	11 t /s on Jets on Orin Nan o; lapt op CPU ≈5-1 0 t/s → 400 toke ns ≈40- 80 s .

Safety classification	Granite Guardian 4.1 8B	CPU	5-15 s	Similar base model; short input → 5-15 s.
Hugging Face Spaces cold start	Free-tier container	Server	120-300 s (2-5 min)	Domain specific Spaces need 2-5 min to wake up.
Langflow orchestration overhead	Langflow graph	Server	+30-150 s	Observed 5x slower than equivalent Lang

		gGr aph flow s.
Total realistic pipeline	≈900-1900 s (15-32 min)	60 s is not achievable end-to-end.

Verdict: A live, cold-start, full-pipeline demo will never finish within 60 seconds. The dominant cost is document parsing (Docling + Vision). Without drastic changes, the team must pre-process or fake the pipeline.

2. Failure Mode Audit – Step by Step

Step 1: Drop Telemetry CSV

- What breaks: The CSV may have missing channels, irregular sampling, or non-standard headers.
- Visible symptom: The TTM model raises a shape mismatch error or silently produces NaN forecasts.
- Mitigation:
 - Validate the CSV against an expected schema.
 - Impute missing channels with zeros or linear interpolation.
 - Resample to a fixed 1 Hz grid before feeding to TTM.

Step 2: Drop FIA Certificate of Adaptations PDF

- What breaks:

- Locked/scanned PDF: Granite-Docling may fail OCR on image-only pages.
- Non-English PDF: The model was trained predominantly on English; French/German legal text may produce garbled JSON.
- Visible symptom: The JSON output is empty, missing section IDs, or contains gibberish.
- Mitigation:
 - Pre-convert the PDF to text using an OCR engine (Tesseract) as fallback.
 - If the COA is always in English, state that assumption upfront.
 - Manually extract the adaptation parameters and store them as a static JSON – the COA rarely changes per driver.

Step 3: Drop Official Timing Sheet PDF

- What breaks: Irregular formatting (merged cells, variable column counts, multiple tables per page) confuses Granite Vision 4.1.
- Visible symptom: The extracted CSV contains missing columns, shifted data, or only partial rows.
- Mitigation:
 - Use a known, fixed layout timing sheet.
 - Pre-parse the PDF with a template-based extractor (e.g., tabula-py) and feed the CSV directly.
 - Strong recommendation: Bypass this step entirely – the timing sheet is static data that can be pre-converted.

Step 4: Paste Written Debrief

- What breaks: Very little can go wrong here; the text is free-form.
- Visible symptom: N/A.
- Mitigation: Truncate to 200 words if the model context window is limited.

Step 5: Click “Run Race Debrief”

- What breaks:
 - Hugging Face Spaces may rate-limit or kill the job if it exceeds 30 min.
 - The browser may time out the HTTP connection.
- Visible symptom: Spinner spins forever; “504 Gateway Timeout” after 60 s.
- Mitigation:
 - Keep the Space “warm” by pinging it before the demo.
 - Show a pre-cached result and only run a trivial live step.

Step 6: Langflow Visual Graph Animation

- What breaks: The UI may lag or fail to render at the screen-recording resolution.
- Visible symptom: Nodes appear blank, transitions freeze, or the graph is cropped.
- Mitigation:
 - Pre-record the Langflow animation and embed it in the video.
 - If live, use a simple flow with large, high-contrast nodes.

Step 7: Output & Safety Stamp

- What breaks:
 - Granite Guardian may false-positive on legitimate engineering language (“reduce brake force by 12%”).
 - The forecast chart may be noisy or implausible (e.g., predicting negative lap times).
- Visible symptom: The report shows “Recommendation REJECTED – safety violation” or a forecast with extreme spikes.
- Mitigation:
 - Adjust Guardian’s threshold or add custom “engineering-domain” criteria.
 - Post-process the forecast with simple clipping (e.g., lap time within $\pm 5\%$ of historical).
 - Always display a human-readable warning: “Forecast is indicative – verify with a professional engineer.”

3. Top 3 Killer Q&A Questions + Ironclad Answers

Q1: “TTM was trained on minutely/hourly data. Your telemetry is 50 Hz. Doesn’t this violate the model’s operating assumption?”

- What the judge is probing: Does the team understand the model’s limitations?
- Ironclad Answer:
“We resample the telemetry to a 1 Hz grid before feeding it to TTM. The 50 Hz data is first down-sampled via simple averaging, which preserves the trend while staying within TTM’s documented sweet spot of minutely/hourly resolutions. The TTM readme explicitly states it supports ‘minutely to hourly’ point forecasting, so our resampled data is fully in-spec.”
- Deck mitigation: Add a “Data Pre-processing” slide that shows the resampling step.

Q2: “Granite-Docling is a general document model. How can you trust it on a legally binding FIA Certificate that directly impacts driver safety?”

- What the judge is probing: Liability and domain-specific reliability.
- Ironclad Answer:
“We treat Granite-Docling as a suggestion engine, not an authority. The extracted adaptation parameters are displayed to the driver alongside a confidence flag. Any recommendation that touches the COA envelope is reviewed by a human coach before implementation. We also maintain a static, manually-verified JSON copy of the driver’s COA – the Docling step is a convenience, not a dependency.”

- Deck mitigation: Add a “Human-in-the-Loop” label and a disclaimer in the UI.

Q3: “You claim 60 seconds end-to-end. I see 4 large models and document parsing. Show me the latency breakdown that adds up to 60 s.”

- What the judge is probing: Honesty and engineering rigor.
 - Ironclad Answer:
“You’re right – a cold-start pipeline does not fit in 60 s. For today’s demo, we have pre-computed the document parsing and forecast steps. The live 60 s window covers only the Langflow orchestration, the coaching report generation, and the Guardian safety check. The full pipeline, from raw files to report, takes approximately 15-20 minutes on this hardware; our product roadmap includes GPU acceleration and edge-side caching to bring that down to under 2 minutes.”
 - Deck mitigation: Never claim 60 s end-to-end in the pitch. Instead, say “60-second interactive demo after data ingestion.”
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4. Live vs. Pre-Recorded Decision

Approach	Risk	Recommendation
Pre-record entire 60 s demo	Lowest risk. No runtime failures. Judges may suspect pre-recording but it’s standard for hackathons.	Preferred for the 3-minute submission video.

Run full pipeline live	Extremely high risk. Docling alone will blow the time budget; HF cold-start will kill the demo.	Do not attempt.
Hybrid (cached intermediate states, live final outputs)	Moderate risk. The Langflow UI can show pre-populated data; the report generation runs live but with a small model.	Best for a live “judge walk-up” demo. Pre-compute the COA JSON, timing-sheet CSV, and TTM forecast. Show the Langflow graph playing back the pre-computed pipeline. Let the judge see the UI load the final report “live.”

Final Verdict:

- For the submission video, pre-record the entire demo.
- For a live booth demo, use the hybrid approach and never promise “60 seconds from scratch.”

5. Fallback Architecture – “PIT WALL Lite”

If on May 26 the team realizes the full Langflow chain won’t work, deploy this stripped-down version:

Original Step	Lite Replacement
Granite-Docling 258M on COA PDF	Replace with a static JSON file containing the driver's adaptations.
Granite Vision 4.1 on timing sheet PDF	Replace with a pre-parsed CSV (manually converted).
Granite-TimeSeries-TTM zero-shot forecast	Keep – it's fast enough (<15 s).
Granite 4.1 8B coaching report	Swap to Granite 4.1 3B Instruct – faster, smaller output.
Granite Guardian 4.1 8B safety check	Swap to Granite-Guardian-HAP-38M for near-instant classification.
Langflow orchestration	Remove Langflow. Use a simple Python script with a static HTML/CSS animation that mimics the graph.

Pitch repositioning: "PIT WALL Lite demonstrates the core AI race-engineer loop – ingest structured data, forecast, report, and safety-stamp – in under 60 seconds. The full version adds automated document understanding for zero-touch operation."

6. The Single Most Important Engineering Decision the Team Hasn't Made Yet

“Should document parsing happen in the critical path of the demo?”

The answer is no. The team is betting the entire demo on two document-understanding models (Docling + Vision) that collectively consume 90% of the latency budget and introduce 80% of the failure modes.

The decision to make now:

Treat the COA and timing sheet as pre-computed static data for the demo.

The live pipeline should start from a structured JSON + CSV, not from raw PDFs. This single choice reduces the demo’s technical risk from “almost certain failure” to “manageable with minor hiccups,” and it frees the team to focus on the value-add steps – forecasting, coaching, and safety – that actually differentiate PIT WALL.

“If you ship the demo with live PDF parsing, you are shipping a demo of document processing, not a demo of an AI race engineer.”

Summary of Brutal Truths:

- The 60-second claim is fiction. Be honest in the pitch.
- Granite-Docling is too slow and too unreliable for a live demo. Kill it in the critical path.
- Langflow adds massive overhead; use a simple animation.
- Pre-record the submission video. Hybrid for live walk-ups.
- The team’s biggest asset is the TTM + Granite Instruct + Guardian pipeline – don’t bury it under document parsing.

Good luck. The difference between \$5,000 and nothing is whether you ship a demo that *works* or a demo that *crashes*.

This response is AI-generated, for reference only.